

```

libname fama 'F:\SAS9.4\SAS_9.4\fama-macbeth';
/*****input
data*****/
PROC IMPORT OUT= FAMA.RAW
DATAFILE= "E:\中财硕士\Fama-Macbeth replication\finalraw.csv" ;
RUN;
proc import out=fama.rf
datafile="E:\中财硕士\Fama-Macbeth replication\finalriskfree.csv";
run;
/*****prepare data
*****/
/*delete void data*/
data fama.raw; /*If we change the content, we'd better change the
dataset's name*/
set fama.raw;
libname fama 'F:\Fama-Macbeth8.5';
/*****input
data*****/
PROC IMPORT OUT= WORK.RAW
DATAFILE= "F:\Fama-Macbeth8.5\finalraw.csv" ;
RUN;
/*stkcd: stock code; Trdmnt: Trade date; Mretwd: stock's monthly
return; Cmretwdeq: market's monthly return*/
proc import out= fama.rf
datafile="F:\Fama-Macbeth8.5\finalriskfree.csv";
run;
/*Trdmnt: Trade date; Monrfret: Monthly risk-free return*/
/*****prepare data
*****/
/*delete void data*/
data fama.raw;
set work.raw;
if Mretwd^=. and Cmretwdeq^=. ; run;
/*count length of each stock's time series as num, save stkcd, num in
table fama.number*/
proc sql noprint;
create table fama.number as
select Stkcd, count(stkcd) as num
from fama.raw
group by stkcd;
quit;
data fama.raw2;
merge fama.raw fama.number;
by stkcd;
run;
/*only keep stocks with more than 19-year data, i.e. 12*19=228 months
*/
data fama.original;
set fama.raw2;
if num >= 228;
run;
/*merge with risk-free rate*/
proc sql;
create table fama.neworiginal as
select original.*, rf.Monrfret
from fama.original full join fama.rf

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on original.Trdmnt=rf.Trdmnt;
quit;
proc sort data=fama.neworiginal;
by stkcd trdmnt;
run;
/***** 为检验CAPM做准备（求原文式（10）的自变量） *****/
%macro all(aa);/*每个时间段（共计4个）分别运算portfolio formation period、
estimation period、testing period*/
data fama.period1a;/*提取portfolio formation period的数据*/
set fama.neworiginal;
if %eval(&aa-6)<=year<=%eval(&aa-3);
run;
proc sort data=fama.period1a;
by stkcd trdmnt;run;
proc reg data=fama.period1a noprint outest=fama.period1b; /*portfolio
formation period 时间序列回归求每只股票beta值*/
model Mretwd=Cmretwdeq;
by Stkcd;
run;
data fama.period1b;
set fama.period1b;
keep Stkcd Cmretwdeq;
run;
proc univariate data=fama.period1b; /*求所有股票beta值的分位数*/
var Cmretwdeq;
output out=fama.a_pct pctlpre=p pctlpts=(5 10 15 20 25 30 35 40 45 50
55 60 65 70 75 80 85 90 95);
run;
data fama.period1c;/*根据beta值由小到大将股票分为20组*/
set fama.period1b;
if _n_=1 then set fama.a_pct;
if Cmretwdeq<=p5 then group1=1;
else if Cmretwdeq<=p10 then group1=2;
else if Cmretwdeq<=p15 then group1=3;
else if Cmretwdeq<=p20 then group1=4;
else if Cmretwdeq<=p25 then group1=5;
else if Cmretwdeq<=p30 then group1=6;
else if Cmretwdeq<=p35 then group1=7;
else if Cmretwdeq<=p40 then group1=8;
else if Cmretwdeq<=p45 then group1=9;
else if Cmretwdeq<=p50 then group1=10;
else if Cmretwdeq<=p55 then group1=11;
else if Cmretwdeq<=p60 then group1=12;
else if Cmretwdeq<=p65 then group1=13;
else if Cmretwdeq<=p70 then group1=14;
else if Cmretwdeq<=p75 then group1=15;
else if Cmretwdeq<=p80 then group1=16;
else if Cmretwdeq<=p85 then group1=17;
else if Cmretwdeq<=p90 then group1=18;
else if Cmretwdeq<=p95 then group1=19;
else group1=20;
run;
data fama.group1a; /*提取每一组的股票代码每组为一个投资组合*/
set fama.period1c;

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keep group1 stkcd;
run;
proc sort data=fama.group1a;
by Stkcd group1;
run;
data fama.group1b; /*分组信息与收益信息相结合*/
merge fama.group1a(in=a) fama.neworiginal;
by Stkcd;
if a;
run;
data fama.group1e1; /*提取每一个投资组合estimation period的股票信息*/
set fama.group1b;
if %eval(&aa-2)<=year<=&aa;
run;
proc sort data=fama.group1e1; /*求投资组合收益率Rp*/
by group1 year trdmnt;
run;
proc means data =fama.group1e1 noprint;
var Mretwd Cmretwdeq;
by group1 year trdmnt;
output out=fama.bb mean=Rp Rm;
run;
Ods Output Summary=fama.Summary; /*求Rp和Rm的标准差*/
Proc Means Data=fama.bb n mean std nmiss;
Var Rp Rm;
by group1;
Run;
Ods Output Close;
data fama.bb;
set fama.bb;
drop _TYPE_ _FREQ_;
run;
proc reg data=fama.bb noprint outest=fama.cc adjrsq; /*回归求投资组合的残差平方根和调整R方*/
model Rp=Rm;
by group1;
run;
proc sort data=fama.group1e1;
by group1 stkcd year trdmnt;
run;
proc reg data= fama.group1e1 noprint outest=fama.group1e2; /*estimation period: 每一投资组合组内每股时间序列回归求个股beta值*/
model Mretwd=Cmretwdeq;
by group1 Stkcd;
run;
data fama.group1e2;
set fama.group1e2;
cmretwdeq2=cmretwdeq*cmretwdeq;
run;
proc means data=fama.group1e2 noprint; /*estimation period: 回归所得的结果投资组合组内求平均得到每一投资组合initial value*/
var Cmretwdeq Cmretwdeq2 _RMSE_;
by group1;
output out=fama.group1v_&aa mean=betap betap2 aveStd;
run;

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proc sql; /*将 _RMSE_, _ADJRSQ_, Rp_StdDev, Rm_StdDev, betap, avestd等信息汇总
到一个table*/
create table fama.oo as
select group1, _RMSE_, _ADJRSQ_
from fama.cc;
quit;
proc sql;
create table fama.pp as
select group1, Rp_StdDev, Rm_StdDev
from fama.summary;
quit;
proc sql;
create table fama.qq as
select group1, betap, avestd
from fama.groupiv_&aa;
quit;
data fama.ss1;
merge fama.oo fama.pp;
by group1;
run;
data fama.ss2;
merge fama.ss1 fama.qq;
by group1;
run;
data fama.ss_&aa ; /*即原文中table2*/
set fama.ss2;
betap_StdDev=_RMSE_/ (sqrt(36) *Rm_StdDev);
ratio=_RMSE_/avestd;
run;
proc sort data=fama.group1b; by group1 stkcd trdmnt; run;
proc reg data= fama.group1b noprint outest=fama.group1t1; /*testing
period aa-aa+1年：每一个投资组合组内每股时间序列回归求个股beta*/
where &aa<=year<=%eval(&aa+1);
model Mretwd=Cmretwdeq;
by group1 Stkcd;
run;
data fama.group1b1; /*提取aa+1年收益率信息*/
set fama.group1b;
where year=%eval(&aa+1);
run;
data fama.x1; set fama.group1t1; /*保留回归分析中的个股beta和均方根误差*/
keep group1 Stkcd Cmretwdeq _RMSE_;
rename Cmretwdeq=coeff;
run;
data fama.group1tmt1; /*与aa+1年月收益率信息整合以排除退市股票*/
merge fama.group1b1 fama.x1(in=id);
by group1 Stkcd;
if id;
run;
proc sort data=fama.group1tmt1;
by group1 year Trdmnt; run;
data fama.group1tmt1;
set fama.group1tmt1;
coeff2=coeff*coeff;
run;

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proc means data=fama.group1tm1 noprint; /*对每月投资组合中所有股票beta求平均得
到aa+1年每月每一投资组合的beta和均方根误差*/
var coeff coeff2 _RMSE_ Mretwd Cmretwdeq Monrfret;
by group1 year Trdmnt;
output out=fama.group_<eval>(&aa+1) mean=betap betap2 aveStd Rp Rm Rf;
run;
proc reg data= fama.group1b noprint outest=fama.group1t2; /*testing
period aa-aa+2年：每一个投资组合组内每股时间序列回归求个股beta*/
where &aa<=year<=<eval>(&aa+2);
model Mretwd=Cmretwdeq;
by group1 Stkcd;
run;
data fama.group1b2; /*提取aa+2年收益率信息*/
set fama.group1b;
where year=<eval>(&aa+2);
run;
data fama.x2; set fama.group1t2; /*保留回归分析中的个股beta和均方根误差*/
keep group1 Stkcd Cmretwdeq _RMSE_;
rename Cmretwdeq=coeff;
run;
data fama.group1tm2; /*与aa+2年月收益率信息整合以排除退市股票*/
merge fama.group1b2 fama.x2(in=id);
by group1 Stkcd;
if id;
run;
proc sort data=fama.group1tm2;
by group1 year Trdmnt;run;
data fama.group1tm2;
set fama.group1tm2;
coeff2=coeff*coeff;
run;
proc means data=fama.group1tm2 noprint; /*对每月投资组合中所有股票beta求平均得
到aa+2年每月每一投资组合的beta和均方根误差*/
var coeff coeff2 _RMSE_ Mretwd Cmretwdeq Monrfret;
by group1 year Trdmnt;
output out=fama.group_<eval>(&aa+2) mean=betap betap2 aveStd Rp Rm Rf;
run;
proc reg data= fama.group1b noprint outest=fama.group1t3; /*testing
period aa-aa+3年：每一个投资组合组内每股时间序列回归求个股beta*/
where &aa<=year<=<eval>(&aa+3);
model Mretwd=Cmretwdeq;
by group1 Stkcd;
run;
data fama.group1b3; /*提取aa+3年收益率信息*/
set fama.group1b;
where year=<eval>(&aa+3);
run;
data fama.x3; set fama.group1t3; /*保留回归分析中的个股beta和均方根误差*/
keep group1 Stkcd Cmretwdeq _RMSE_;
rename Cmretwdeq=coeff;
run;
data fama.group1tm3; /*与aa+3年月收益率信息整合以排除退市股票*/
merge fama.group1b3 fama.x3(in=id);
by group1 Stkcd;

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if id;
run;
proc sort data=fama.group1tm3;
by group1 year Trdmnt;run;
data fama.group1tm3;
set fama.group1tm3;
coeff2=coeff*coeff;
run;
proc means data=fama.group1tm3 noprint; /*对每月投资组合中所有股票beta求平均得
到aa+3年每月某一投资组合的beta和均方根误差*/
var coeff coeff2 _RMSE_ Mretwd Cmretwdeq Monrfret;
by group1 year Trdmnt;
output out=fama.group_<eval(&aa+3) mean=betap betap2 aveStd Rp Rm Rf;
run;
%mend;
<macro pp; /*上述宏运行四个时间段*/
do nn=2003 %to 2012 %by 3;
all(&nn);
end;
%mend;
pp;
/*****检验CAPM模型*****/
data fama.pltest; /*将时间段1的estimation period和三个testing periods整合*/
set fama.groupiv_2003 fama.Group_2004 fama.Group_2005 fama.Group_2006;
run;
proc sort data=fama.pltest; by Year Trdmnt; run;
data fama.p2test; /*将时间段2的estimation period和三个testing periods整合*/
set fama.groupiv_2006 fama.Group_2007 fama.Group_2008 fama.Group_2009;
run;
proc sort data=fama.p2test; by Year Trdmnt; run;
data fama.p3test; /*将时间段3的estimation period和三个testing periods整合*/
set fama.groupiv_2009 fama.Group_2010 fama.Group_2011 fama.Group_2012;
run;
proc sort data=fama.p3test; by Year Trdmnt; run;
data fama.p4test; /*将时间段4的estimation period和三个testing periods整合*/
set fama.groupiv_2012 fama.Group_2013 fama.Group_2014 fama.Group_2015;
run;
proc sort data=fama.p4test; by Year Trdmnt; run;
macro capm(m);
data fama.p&m.test; /*当前一期收益率与之前一期的beta、beta平方项和均方根误差相对
应*/
set fama.p&m.test;
betap_1=lag20(betap);
aveStd_1=lag20(aveStd);
betap2_1=lag20(betap2);
drop _FREQ_ _TYPE_;
run;
data fama.p&m.testn;
set fama.p&m.test;
if year^=.;
run;
proc reg data=fama.p&m.testn outest=fama.p&m.resulta tableout
ADJRSQ; /*panela*/
Model Rp=betap_1;

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by Trdmnt;
quit;
proc reg data=fama.p&m.testn outest=fama.p&m.resultb tableout
ADJRSQ; /*panelb*/
Model Rp=betap_1 betap2_1;
by Trdmnt;
quit;
proc reg data=fama.p&m.testn outest=fama.p&m.resultc tableout
ADJRSQ; /*panelc*/
Model Rp=betap_1 aveStd_1;
by Trdmnt;
quit;
proc reg data=fama.p&m.testn outest=fama.p&m.resultd tableout
ADJRSQ; /*paneld*/
Model Rp=betap_1 betap2_1 aveStd_1;
by Trdmnt;
quit;

data fama.result&m.a; /*提取回归系数*/
set fama.p&m.resulta;
if _TYPE_='PARMS';
by trdmnt;
run;
proc sql; create table fama.yyy&m.a as
select distinct result&m.a.*, rm, rf
from fama.result&m.a left join fama.p&m.test on
result&m.a.trdmnt=p&m.test.trdmnt;
quit;
data fama.yyy&m.a; set fama.yyy&m.a; diff1=intercept-rf; diff2=rm-
rf; run; /*计算截距项、市场收益率和无风险利率的差值分别记为diff1和diff2*/
data fama.stder&m.a; /*提取回归系数的标准误*/
set fama.P&m.resulta;
if _TYPE_='STDERR';
by Trdmnt;
drop _MODEL_ _DEPVAR_ _RMSE_ Rp;
run;
proc univariate data=fama.yyy&m.a; /*计算回归系数、R方、Rm、Rf和diff的均值、标
准差、方差和t值*/
var Intercept betap_1 _ADJRSQ_ Rm Rf diff1 diff2;
output out=fama.table1_&m._a
MEAN=meanr0 meanr1 meanRR meanrm meanrf meandif1 meandif2
STD=stdr0 stdr1 stdRR stdrm stdrf stddif1 stddif2
VAR=varr0 varr1 varRR varrm varrf vardif1 vardif2
t=tr0 tr1 trr trm trf tdif1 tdif2;
quit;
data fama.stder&m.a;
set fama.stder&m.a;
intercept2=intercept*intercept;
betap_12=betap_1*betap_1;
run;
Ods Output Means.Summary=fama.table2_&m._a; /*求回归系数标准误的平方的均值*/
Proc Means Data=fama.stder&m.a mean ;
Var Intercept2 betap_12;
Run;
Ods Output Close;

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data fama.result&m.b; /*提取回归系数*/
set fama.p&m.resultb;
if _TYPE_='PARMS';
by trdmnt;
run;
proc sql;create table fama.yyy&m.b as
select distinct result&m.b.*,rm,rf
from fama.result&m.b left join fama.p&m.test on
result&m.b.trdmnt=p&m.test.trdmnt;
quit;
data fama.yyy&m.b;set fama.yyy&m.b;diff1=intercept-rf;diff2=rm-
rf;run; /*计算截距项、市场收益率和无风险利率的差值分别记为diff1和diff2*/
data fama.stderr&m.b; /*提取回归系数的标准误*/
set fama.P&m.resultb;
if _TYPE_='STDERR';
by Trdmnt;
drop _MODEL_ _DEPVAR_ _RMSE_ Rp;
run;
proc univariate data=fama.yyy&m.b; /*计算回归系数、R方、Rm、Rf和diff的均值、标
准差、方差和t值*/
var Intercept betap_1 betap2_1 _ADJRSQ_ Rm Rf diff1 diff2;
output out=fama.table1_&m._b
MEAN=meanr0 meanr1 meanr2 meanRR meanrm meanrf meandif1 meandif2
STD=stdr0 stdr1 stdr2 stdRR stdrm stdrf stddif1 stddif2
VAR=varr0 varr1 varr2 varRR varrm varrf vardif1 vardif2
t=tr0 tr1 tr2 trr trm trf tdif1 tdif2;
quit;
data fama.stderr&m.b;
set fama.stderr&m.b;
intercept2=intercept*intercept;
betap_12=betap_1*betap_1;
betap2_12=betap2_1*betap2_1;
run;
Ods Output Means.Summary=fama.table2_&m._b; /*求回归系数标准误的平方的均值*/
Proc Means Data=fama.stderr&m.b mean ;
Var Intercept2 betap_12 betap2_12;
Run;
Ods Output Close;

data fama.result&m.c; /*提取回归系数*/
set fama.p&m.resultc;
if _TYPE_='PARMS';
by trdmnt;
run;
proc sql;create table fama.yyy&m.c as
select distinct result&m.c.*,rm,rf
from fama.result&m.c left join fama.p&m.test on
result&m.c.trdmnt=p&m.test.trdmnt;
quit;
data fama.yyy&m.c;set fama.yyy&m.c;diff1=intercept-rf;diff2=rm-
rf;run; /*计算截距项、市场收益率和无风险利率的差值分别记为diff1和diff2*/
data fama.stderr&m.c; /*提取回归系数的标准误*/
set fama.P&m.resultc;
if _TYPE_='STDERR';
by Trdmnt;

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drop _MODEL_ _DEPVAR_ _RMSE_ Rp;
run;
proc univariate data=fama.yyy&m.c; /*计算回归系数、R方、Rm、Rf和diff的均值、标准差、方差和t值*/
var Intercept betap_1 aveStd_1 _ADJRSQ_ Rm Rf diff1 diff2;
output out=fama.table1_&m._c
MEAN=meanr0 meanr1 meanr3 meanRR meanrm meanrf meandif1 meandif2
STD=stdr0 stdr1 stdr3 stdRR stdrm stdrf stddif1 stddif2
VAR=varr0 varr1 varr3 varRR varrm varrf vardif1 vardif2
t=tr0 tr1 tr3 trr trm trf tdif1 tdif2;
quit;
data fama.stderr&m.c;
set fama.stderr&m.c;
intercept2=intercept*intercept;
betap_12=betap_1*betap_1;
aveStd_12=aveStd_1*aveStd_1;
run;
Ods Output Means.Summary=fama.table2_&m._c; /*求回归系数标准误的平方的均值*/
Proc Means Data=fama.stderr&m.c mean ;
Var Intercept2 betap_12 aveStd_12;
Run;
Ods Output Close;

data fama.result&m.d; /*提取回归系数*/
set fama.p&m.resultd;
if _TYPE_='PARMS';
by trdmnt;
run;
proc sql; create table fama.yyy&m.d as
select distinct result&m.d.*, rm, rf
from fama.result&m.d left join fama.p&m.test on
result&m.d.trdmnt=p&m.test.trdmnt;
quit;
data fama.yyy&m.d; set fama.yyy&m.d; diff1=intercept-rf; diff2=rm-rf; run; /*计算截距项、市场收益率和无风险利率的差值分别记为diff1和diff2*/
data fama.stderr&m.d; /*提取回归系数的标准误*/
set fama.P&m.resultd;
if _TYPE_='STDERR';
by Trdmnt;
drop _MODEL_ _DEPVAR_ _RMSE_ Rp;
run;
proc univariate data=fama.yyy&m.d; /*计算回归系数、R方、Rm、Rf和diff的均值、标准差、方差和t值*/
var Intercept betap_1 betap2_1 aveStd_1 _ADJRSQ_ Rm Rf diff1 diff2;
output out=fama.table1_&m._d
MEAN=meanr0 meanr1 meanr2 meanr3 meanRR meanrm meanrf meandif1 meandif2
STD=stdr0 stdr1 stdr2 stdr3 stdRR stdrm stdrf stddif1 stddif2
VAR=varr0 varr1 varr2 varr3 varRR varrm varrf vardif1 vardif2
t=tr0 tr1 tr2 tr3 trr trm trf tdif1 tdif2;
quit;
data fama.stderr&m.d;
set fama.stderr&m.d;
intercept2=intercept*intercept;
betap_12=betap_1*betap_1;
betap2_12=betap2_1*betap2_1;

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aveStd_12=aveStd_1*aveStd_1;
run;
Ods Output Means.Summary=fama.table2_&m._d; /*求回归系数标准误的平方的均值*/
Proc Means Data=fama.stderr&m.d mean ;
Var Intercept2 betap_12 betap2_12 aveStd_12;Run;
Ods Output Close;
%mend;
%capm(1);
%capm(2);
%capm(3);
%capm(4);
ods html file="E:\中财硕士\Fama-Macbeth replication\table1body.html.";
data fama.table1; /*即原文中table3*/
set fama.table1_1_a fama.table1_2_a fama.table1_3_a fama.table1_4_a
fama.table1_1_b fama.table1_2_b fama.table1_3_b fama.table1_4_b
fama.table1_1_c fama.table1_2_c fama.table1_3_c fama.table1_4_c
fama.table1_1_d fama.table1_2_d fama.table1_3_d fama.table1_4_d;
run;
proc print;
run;
ods html close;
ods html file="E:\中财硕士\Fama-Macbeth replication\table2body.html.";
data fama.table2; /*即原文中table4*/
set fama.table2_1_a fama.table2_2_a fama.table2_3_a fama.table2_4_a
fama.table2_1_b fama.table2_2_b fama.table2_3_b fama.table2_4_b
fama.table2_1_c fama.table2_2_c fama.table2_3_c fama.table2_4_c
fama.table2_1_d fama.table2_2_d fama.table2_3_d fama.table2_4_d;
run;
proc print;
run;
ods html close;

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